6. Optionals and Optional Handling

Task 1:

Code:

```
struct Address {
    let city: String?
}
struct User {
    let name: String
    let age: Int?
    let address: Address?
}
func describe(_ user: User) -> String {
    guard let age = user.age else {
        return "\(user.name): Age is unknown"
    }
    let city = user.address?.city ?? "unknown city"
    return "\(user.name), Age: \(age), City: \(city)"
}
let user1 = User(
    name: "Asha",
    age: 22,
    address: Address(city: "Chennai")
)
let user2 = User(
    name: "Ravi",
    age: nil,
    address: Address(city: "Coimbatore")
)
```

```
let user3 = User(
    name: "Kiran",
    age: 21,
    address: Address(city: nil)
)
print(describe(user1))
print(describe(user2))
print(describe(user3))
let num1 = Int("42")
let num2 = Int("abc")
print(num1 as Any)
print(num2 as Any)
```

Output:

```
Output

Asha, Age: 22, City: Chennai
Ravi: Age is unknown
Kiran, Age: 21, City: unknown city
Optional(42)
nil
```

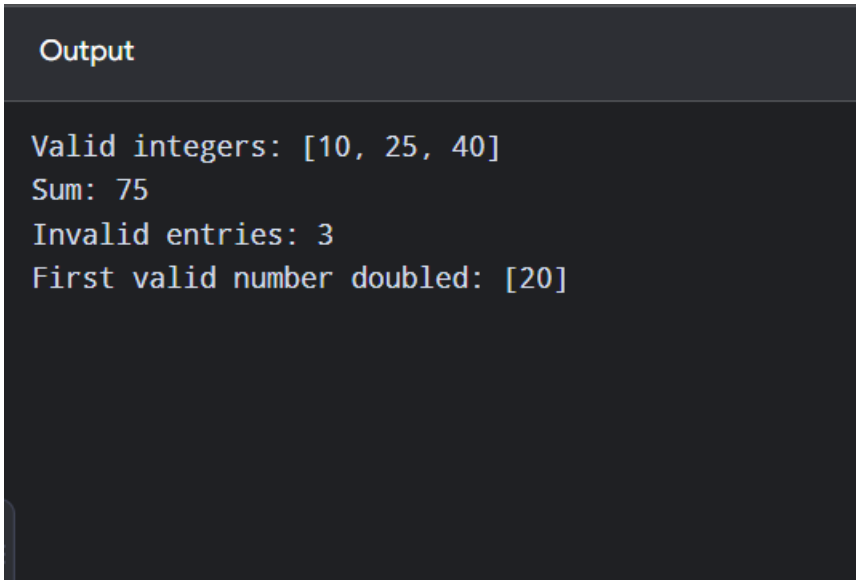
Task 2:

Code:

```
let values = ["10", "x", "25", "", "7.5", "40"]
let validNumbers = values.compactMap { Int($0) }
```

```
print("Valid integers:", validNumbers)
let sum = validNumbers.reduce(0, +)
print("Sum:", sum)
let invalidCount = values.count - validNumbers.count
print("Invalid entries:", invalidCount)
let doubled = values.lazy
    .compactMap { Int($0) }
    .prefix(1)
    .map { $0 * 2 }
print("First valid number doubled:", Array(doubled))
```

Output:



Task 3:

Code:

```
let scores = ["Asha": 90, "Ravi": 72]
let kiran = scores["Kiran"]
let number = Int("12a")
if let score = kiran {
    if let value = number {
```

```
        print(score + value)
    } else {
        print("Invalid number: Cannot convert 12a to an integer")
    }
} else {
    print("Score not found for Kiran")
}
```

Output:

